

LIFE PROCESSES

Introduction :

- How do we know something is alive?
 - Living beings show **life activities**.
 - Even while sleeping, animals are alive because they **breathe**.



Dog



Cow



Man

Examples of living beings

All are alive because they breathe.



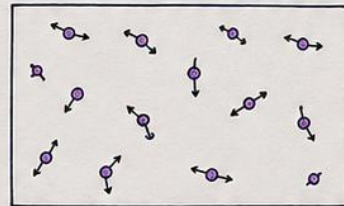
- Are **plants** alive?

- Plants are **green** but being green is not the only reason.
- Plants **grow** with time → **movement** due to growth shows **life**.

- Is movement the sign of life?

- **Visible movement** is not the correct identification of life.
- Every living organism has molecular movement inside its cells (cannot be seen by naked eye).

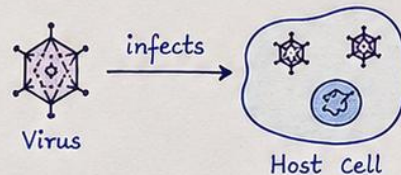
Molecular Movement



Molecules are constantly moving in all directions.

- What about **viruses**?

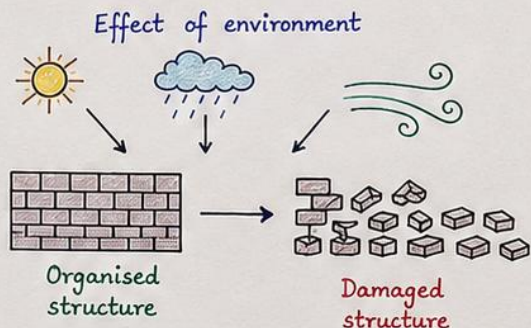
- No molecular movement outside host cell.
- Become active only after **infecting** a cell.
- Hence, debate: are viruses alive or not?



Virus becomes active only inside a cell.

- Why are molecular movements needed?

- Living organisms are highly **organised**.
- Body has **tissues** → cells → smaller parts.
- Environment **damages** this organised structure.
- If order is completely disturbed, organism is **no longer alive**.
- Organism must **repair & maintain** its body.
- Body structures are made of **molecules**.
- Molecules move inside the body to **repair** and **maintain** structures.



Life Processes = Processes that help to repair, maintain and keep the body functioning.

→ These maintenance activities are called Life Processes.